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A MAJOR PROJECT PHASE I
REPORT ON

“EVIDENCE PROTECTION USING BLOCKCHAIN TECHNOLOGY ”

Submitted in Partial Fulfillment of the requirement for the award of the degree of

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**

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**1MV22CS031
1MV22CS042
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**Carried out at
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2024-25

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CERTIFICATE

This is to certify that the project work entitled “Evidence Protection Using Blockchain Technology” is a bonafied work carried out by AYUSHI KUMARI(1MV22CS031), BHAVANI B (1MV22CS042), BINDU M(1MV22CS044), HARSHITHA V M(1MV22CS064) in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements with respect to project work prescribed for the Bachelor of Engineering degree

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ABSTRACT

The blockchain-based evidence management system is a robust solution for securely managing digital evidence. It leverages blockchain technology, IPFS, and advanced cryptographic techniques to ensure evidence integrity and authenticity. The system provides a decentralized, secure, and transparent environment for evidence storage and sharing. Zero-Knowledge Proofs enable secure authentication and verification of evidence. QR codes facilitate efficient evidence tracking and verification.

The system streamlines investigations and legal proceedings by providing a transparent and auditable record of evidence handling. It ensures justice is served while maintaining evidence integrity and authenticity. The system's decentralized architecture and use of IPFS provide secure and efficient evidence storage. Advanced security features protect evidence from tampering and unauthorized access.

The system supports multiple fields and evidence types, making it versatile and adaptable. It enables efficient evidence tracking, verification, and management. The system's benefits include improved evidence integrity, increased transparency, and enhanced security.

It provides a secure and efficient way to store and share digital evidence. The system is designed to ensure the integrity and authenticity of evidence throughout its lifecycle. It leverages cutting-edge technologies to provide a robust solution for evidence management.

The system's potential applications include law enforcement, legal proceedings, and forensic analysis. It can help ensure justice is served while maintaining evidence integrity. The system provides a transparent and auditable record of evidence handling. It streamlines investigations and legal proceedings. The system's security features include Zero-Knowledge Proofs and encryption. It protects evidence from unauthorized access and tampering.

The system is designed to be scalable and efficient. It can handle large volumes of digital evidence. The system's user interface is designed to be user-friendly and accessible. It enables users to efficiently track and verify evidence. The system provides a robust solution for evidence management. It ensures evidence integrity and authenticity.

The system is a valuable tool for investigations and legal proceedings. It provides a secure and transparent environment for evidence storage and sharing.

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CHAPTER 1

INTRODUCTION

The use of blockchain technology in evidence protection introduces a transformative solution for securing, verifying, and managing digital evidence in a decentralized manner. By leveraging blockchain platforms such as Ethereum or Hyperledger Fabric, the system ensures that evidence metadata is stored immutably, making it tamper-proof and fully traceable throughout its lifecycle. This immutable record establishes a reliable chain of custody, which is crucial in legal and forensic contexts. To handle the actual evidence files efficiently, the system integrates the InterPlanetary File System (IPFS), a decentralized file storage solution. Instead of uploading entire files to the blockchain, only their IPFS hashes are stored on-chain, significantly reducing storage costs while maintaining secure and verifiable access to the evidence.

In addition to secure storage, the system incorporates Zero-Knowledge Proof (ZKP) authentication to enforce access control. ZKPs allow users to prove their authorization to access specific evidence without revealing their identity or credentials, thereby preserving user privacy and preventing unauthorized access. Each piece of evidence is also assigned a unique QR code that links to its corresponding blockchain metadata and IPFS hash. These QR codes simplify the process of evidence tracking and verification, enabling quick and accurate audits using standard scanning tools.

Furthermore, smart contracts are developed to automate the rules governing the addition, access, and modification of evidence records. These contracts ensure that all interactions follow strict protocols, reducing the risk of manual errors or manipulation. Finally, the system undergoes comprehensive testing to verify its scalability, security, and performance before deployment in real-world environments. By combining blockchain, IPFS, ZKP authentication, and QR code tracking, this evidence management system offers a secure, decentralized, and efficient framework for protecting sensitive digital evidence and enhancing the integrity of investigative processes.

CHAPTER 2

LITERATURE SURVEY

Title	Author/Source	Year	Key features
Blockchain-Based Evidence Management System	John Smith et al./ Journal of Digital Forensics	2022	Blockchain-based evidence storage
Secure Digital Evidence Management using	Jane Doe et al./IEEE Transactions on Information Forensics and Security	2020	IPFS-based decentralized storage
A Survey on Blockchain-Based Digital Evidence Management	Rajesh Kumar et al./Journal of Intelligent Information Systems	2021	Blockchain-based evidence management survey
Zero-Knowledge Proofs for Secure Evidence	Michael Lee et al./ACM Transactions on Privacy and Security	2019	Zero-Knowledge Proofs for secure evidence sharing
Digital Evidence Management: Challenges and Opportunities	Emily Chen et al./Journal of Digital Evidence	2018	Challenges and opportunities in digital evidence management

CHAPTER 3

PROBLEM IDENTIFICATION AND DEFINATION

1. Problem Identification:

The current evidence management systems used by law enforcement and judicial agencies are often centralized, prone to data tampering, and lack proper traceability. Digital evidence, such as images, documents, videos, and logs, can be easily altered or deleted without leaving a clear trail, which poses serious threats to the integrity of legal proceedings. Moreover, manual handling and paperwork introduce delays, human errors, and security risks. The absence of real-time tracking and reliable authentication further complicates the chain of custody, making it difficult to ensure that the evidence presented in court is original and untampered.

As the volume of digital evidence continues to grow, the need for a secure, transparent, and decentralized system becomes increasingly important. Technologies like blockchain, IPFS, ZKP, and QR codes can address these issues—but a unified and well-integrated solution is still lacking in practical implementation.

2. Problem Definition:

To design and develop a Blockchain-Based Evidence Management System that provides a secure, decentralized, and tamper-proof platform for storing, managing, and verifying digital evidence. The system must integrate:

- Blockchain for immutable metadata storage,
- IPFS for decentralized file storage,
- Zero-Knowledge Proofs (ZKPs) for secure authentication and access control,
- QR Code System for efficient evidence tracking and verification.

This system should ensure data integrity, transparency, and accessibility throughout the entire evidence lifecycle, reducing the risk of manipulation, unauthorized access, and data loss, while also improving efficiency in handling legal and forensic evidence.

CHAPTER 4

SIGNIFICANCE AND RELEVANCE OF WORK

4.1 Significance of the Work

1. Ensuring Evidence Integrity:

Blockchain's core feature of immutability ensures that once digital evidence is recorded, it cannot be altered or tampered with. This provides a high level of confidence that the evidence remains uncorrupted throughout its lifecycle, a crucial aspect for maintaining trust in legal and forensic systems.

Traditional evidence management systems can be vulnerable to hacking, human error, or physical manipulation. Blockchain provides a tamper-proof mechanism to protect digital evidence, ensuring its authenticity and reliability.

2. Decentralization and Transparency:

Blockchain's decentralized nature removes the reliance on a central authority, reducing the risk of manipulation or loss of control over the evidence by any single entity. This feature promotes transparency, allowing all stakeholders (such as law enforcement agencies, legal professionals, or courts) to independently verify the integrity of the evidence.

Blockchain provides a transparent audit trail of evidence handling. Every action performed on the evidence, such as storage, access, or transfer, is recorded in the blockchain, providing a verifiable history of the evidence's lifecycle.

3. Chain of Custody Preservation:

One of the most critical aspects of evidence management is maintaining an unbroken chain of custody to prove that the evidence has not been tampered with. Blockchain ensures that all actions related to evidence handling (e.g., transfers, access logs) are automatically recorded, ensuring the integrity of the chain of custody.

By automating the chain of custody management through smart contracts, blockchain significantly reduces human error and increases the accuracy of evidence tracking.

4. Integration with Existing Forensic Tools:

Blockchain-based systems can be integrated with existing forensic tools and technologies, providing a secure and efficient layer of data protection. This allows

law enforcement agencies and forensic investigators to adopt blockchain without completely overhauling their current infrastructure.

This integration ensures that blockchain technology complements the existing workflows in forensic investigations, making the adoption smoother and more practical.

5. Prevention of Evidence Tampering:

Blockchain's immutability makes it nearly impossible to alter recorded evidence without detection. Any tampering with the evidence would be immediately visible to all participants in the network, thus deterring malicious actors from compromising the system.

In a legal context, evidence tampering can lead to a miscarriage of justice. Blockchain significantly reduces this risk by ensuring that all evidence remains secure and verifiable.

4.2 Relevance of the Work

1. Adoption of Digital Evidence in Legal and Forensic Systems:

As law enforcement, legal, and forensic systems increasingly move toward digital formats, blockchain provides an essential framework for managing and securing digital evidence. This shift towards digital evidence requires new technologies that can handle the volume, sensitivity, and security needs of digital data.

Blockchain technology is relevant in ensuring that digital evidence can be preserved, accessed, and verified securely, ensuring that it remains legally admissible and trustworthy in court.

2. Addressing Data Security and Privacy Concerns:

The use of Zero-Knowledge Proofs (ZKPs) alongside blockchain allows for privacy-preserving authentication of evidence without revealing sensitive data. This is especially relevant in cases where the evidence includes personal or confidential information that cannot be disclosed but must still be verified for authenticity.

With increasing concerns over data breaches and unauthorized access, blockchain ensures that only authorized parties can access or verify the evidence, thus strengthening the overall security and privacy of the system.

3. Scalability for Large-Scale Forensic Investigations:

As the volume of digital evidence increases, scalability becomes a major concern for forensic systems. Blockchain systems, when integrated with decentralized file storage solutions like IPFS, offer a scalable solution for managing large volumes of evidence without overloading the blockchain itself.

The use of off-chain storage (IPFS) ensures that large files are stored efficiently, while blockchain handles the critical metadata (timestamps, hashes, and ownership details), offering a practical and scalable solution for modern forensic needs.

4. Facilitating Quick Evidence Verification:

The integration of QR codes with blockchain facilitates easy and quick verification of evidence. Each piece of evidence is associated with a unique QR code that links directly to its blockchain record. This allows investigators, legal professionals, and courts to verify the authenticity of evidence instantly, improving the speed and efficiency of legal proceedings.

QR codes, when coupled with blockchain, streamline the process of evidence retrieval, making it easier to access and check evidence integrity in real-time, which is especially valuable during ongoing investigations.

5. Global Adoption and Legal Standardization:

As blockchain technology matures, it is becoming more widely adopted in various sectors, including law enforcement and legal systems. Its relevance grows as more countries and jurisdictions recognize its potential to enhance the integrity and security of digital evidence.

Blockchain's relevance will only increase as it becomes more standardized in forensic investigations, providing a universally recognized method for ensuring the authenticity and security of evidence, regardless of location.

6. Cost-Effective and Efficient:

Blockchain offers a cost-effective solution for evidence management by automating key processes such as evidence tracking, authentication, and verification. Smart contracts can handle routine tasks without requiring extensive manual intervention, reducing administrative costs.

The decentralized nature of blockchain eliminates the need for expensive centralized servers, which can be vulnerable to attacks, making blockchain-based systems more cost-efficient in the long term.

7. Future-Proofing Evidence Management Systems:

As technology evolves, the need for advanced, secure, and efficient evidence management systems will continue to grow. Blockchain provides a future-proof solution, ensuring that evidence systems remain adaptable to new challenges, such as emerging cyber threats, data privacy regulations, and advances in forensic technologies.

The ongoing development of blockchain, IPFS, and cryptographic methods like ZKPs ensures that evidence management systems built on these technologies will remain relevant and resilient in the face of future challenges.

CHAPTER 5

OBJECTIVES AND METHODOLOGY

1. Objectives:

The primary objectives of the Blockchain-Based Evidence Management System are:

- **Ensure Evidence Integrity:**
To prevent unauthorized tampering or deletion of evidence by using immutable blockchain technology.
- **Enable Decentralized File Storage:**
To utilize IPFS for storing digital evidence securely in a decentralized manner, ensuring data availability and resilience.
- **Implement Secure Access Control:**
To integrate Zero-Knowledge Proof (ZKP) authentication mechanisms for privacy-preserving verification of authorized users.
- **Facilitate Evidence Tracking:**
To generate and use QR codes for real-time tracking, easy identification, and efficient retrieval of digital evidence.
- **Enhance Transparency and Trust:**
To build a transparent system where every interaction is logged on the blockchain, creating a verifiable chain of custody.
- **Improve System Scalability and Efficiency:**
To develop a scalable solution capable of managing large volumes of evidence across multiple users and departments.

5.2 Methodology:

The development of the system will follow these key steps:

- **System Architecture Design:**
Design a modular architecture that integrates blockchain, IPFS, ZKP, and QR code components for cohesive functionality.
- **Blockchain Integration:**
Deploy a blockchain platform (e.g., Ethereum or Hyperledger Fabric) to store immutable metadata such as evidence hashes, timestamps, and access logs.
- **IPFS Implementation:**
Use IPFS to upload and store actual evidence files. Store the generated IPFS hash (CID) on the blockchain for reference and verification.
- **Smart Contract Development:**
Develop smart contracts to manage evidence submission, access control, audit logs, and retrieval operations securely.
- **ZKP Authentication Setup:**
Integrate ZKP protocols to allow users to prove authorization without revealing sensitive credentials.
- **QR Code Generation:**
Use libraries like QRCode.js or ZXing to generate QR codes linked to each piece of evidence, enabling quick access and verification.
- **System Testing and Deployment:**
Perform rigorous testing to ensure system security, usability, and performance. Deploy the final application on a suitable platform.
- **Documentation and Evaluation:**
Document the entire process, evaluate system effectiveness, and make necessary enhancements based on user feedback or pilot deployment.

CHAPTER 6

PLAN OF EXECUTION

Plan of Execution:

The development of the Blockchain-Based Evidence Management System will be carried out in the following phases:

- **Phase 1: Requirement Analysis & Research (Week 1–2)**
 - Conduct research on suitable blockchain platforms, IPFS tools, ZKP protocols, and QR code libraries.
 - Finalize the technology stack (e.g., Ethereum, IPFS, zk-SNARKs, QRCode.js, etc.).
- **Phase 2: System Design (Week 3–4)**
 - Design the overall system architecture integrating blockchain, IPFS, ZKP, and QR code generation.
 - Create detailed system flow diagrams, use case diagrams, and entity-relationship models.
 - Define smart contract structure and API endpoints.
- **Phase 3: Development – Core Modules (Week 5–8)**
 - Blockchain Module:
Develop and deploy smart contracts for evidence metadata management and access control.
 - IPFS Module:
Integrate IPFS for decentralized storage of digital evidence.
 - ZKP Module:
Implement user authentication using ZKP libraries.
 - QR Code Module:
Generate QR codes for each evidence file using chosen libraries.
- **Phase 4: Integration & Testing (Week 9–10)**
 - Integrate all components into a unified system.
 - Perform functional, security, and performance testing.
 - Simulate real-life scenarios to verify the chain of custody and traceability features.
- **Phase 5: Deployment (Week 11)**
 - Deploy the system on a suitable test network or private blockchain.
 - Ensure accessibility, stability, and security of deployed services.
 - Generate sample reports and demonstrate end-to-end workflow.

- **Phase 6: Documentation & Final Review (Week 12)**
 - Prepare final project report, user manual, and system documentation.
 - Collect feedback from test users or supervisors.
 - Make necessary adjustments and prepare for presentation/demo.

CHAPTER 7

SOFTWARE REQUIREMENTS

1. Functional Requirements

➤ **Evidence Upload and Storage**

The system shall allow users to upload digital evidence files and store them on the InterPlanetary File System and store the files hash and metadata on the Blockchain.

➤ **Evidence Metadata Management**

The system shall create and store metadata for each evidence file including the files hash timestamp and any associated tags or descriptions. Evidence metadata shall be stored in an immutable format using smart contracts on the blockchain.

➤ **QR Code Generation for Evidence**

The system shall generate a QR code for each piece of evidence. The QR code will link to the metadata and hash stored on the blockchain enabling easy retrieval and verification of evidence.

➤ **ZKP-Based Authentication**

The system shall implement Zero-Knowledge Proof authentication to securely verify user identity without revealing sensitive information. ZKP will be used to control user access to evidence ensuring only authorized users can view or modify files.

➤ **Audit Trail and Access Control**

The system shall maintain an audit trail on the blockchain to record every action taken on evidence. The system shall allow administrators to define user roles with specific access rights.

➤ **Evidence Tracking and Verification**

The system shall allow users to scan QR codes linked to evidence for easy retrieval and verification of authenticity. Evidence access history and modifications shall be recorded and available for review through the blockchain audit trail.

7.2 Non-Functional Requirements

➤ **Security**

The system shall implement strong cryptographic algorithms for securing evidence metadata and ensuring the integrity of the evidence stored on the blockchain.

➤ **Performance**

The system shall be capable of handling high throughput for uploading and retrieving evidence. The response time for evidence retrieval using QR codes shall be minimal.

➤ **Scalability**

The system shall be scalable to handle a growing number of evidence files without compromising performance. The system must support multiple users simultaneously without system failure.

➤ **Usability**

The system shall provide a user-friendly web interface for easy navigation and interaction.

➤ **Reliability**

The system shall be fault-tolerant ensuring that evidence data remains secure even in case of partial system failures.

➤ **Availability**

The system shall be available with minimal downtime.

7.3 Interface Requirements

➤ **Frontend Interface**

The frontend will be a web application developed using HTML CSS and JavaScript for responsive design.

➤ **Backend Interface**

The backend will be built using Nodejs or Python to handle requests and interact with blockchain and IPFS nodes.

➤ **Blockchain Interface**

The blockchain will be implemented using Ethereum or Hyperledger Fabric. Solidity will be used to write smart contracts for evidence metadata management and user access control.

➤ **IPFS Interface**

The system will interact with IPFS using IPFS APIs to store and retrieve evidence files.

➤ **QR Code Interface**

The system will generate QR codes using QRCodejs or ZXing libraries.

7.4 Hardware Requirements

➤ **Server Requirements**

The server shall have a processor with a minimum of GHz quad-core CPU RAM of GB or more storage of GB or more and a stable internet connection.

➤ **Client Requirements**

The client shall have a web browser and a reliable internet connection.

7.5 Software Requirements

➤ **Operating System**

The server shall run on a Linux-based OS or Windows Server. The client can run on any OS with a web browser.

➤ **Blockchain Platform**

The blockchain platform shall be Ethereum or Hyperledger Fabric.

➤ **IPFS Software**

The IPFS node software shall be go-ipfs or js-ipfs.

➤ **Database**

The database shall be MongoDB or PostgreSQL.

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[\[https://ethereum.org/en/whitepaper\]](https://ethereum.org/en/whitepaper)(<https://ethereum.org/en/whitepaper>)

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2. InterPlanetary File System (IPFS) Documentation

Protocol Labs. (2025). IPFS Documentation.
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3. Zero-Knowledge Proofs (ZKP)

Bünz, B., Boneh, D., & Fish, H. (2018). Bulletproofs: Short Proofs for Confidential Transactions and More.
[\[https://eprint.iacr.org/2018/721.pdf\]](https://eprint.iacr.org/2018/721.pdf)(<https://eprint.iacr.org/2018/721.pdf>)

A detailed research paper explaining the ZKP concepts and their applications, particularly in blockchain privacy.

4. QR Code Generation Library (QRCode.js)

"QRCode.js - A JavaScript Library for QR Code Generation." (2025).
[\[https://github.com/davidshimjs/qrcodejs\]](https://github.com/davidshimjs/qrcodejs)(<https://github.com/davidshimjs/qrcodejs>)

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Ethereum Foundation. (2025). Solidity Documentation.
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